

Towards the Design of a Wearable Multi-modal Haptic Device to Study Interaction Effects in Haptic Perception*

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ABSTRACT

Multi-modal haptic devices have the capability to convey different types of sensory information through the mechanoreceptors in the skin, allowing for a more accurate and realistic perception of haptic feedback. However, integrating different types of actuators into a device is difficult as it can lead to perceptual interference and potential masking effects. This work outlines the design of a wearable device to study how haptic perception thresholds vary in the presence of more than one type of actuation system in a wearable device, specifically focusing on compression and vibrotactile haptic feedback. The paper also highlights the efforts so far in terms of prototype development and a proposed user study to evaluate interference/interaction between the different haptic stimuli.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Haptic devices.

KEYWORDS

Haptic Devices, Sensory Perception, Compression Feedback, Wearable Technology, Soft Robotics

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1 INTRODUCTION

Haptic feedback has applications ranging from sensory substitution feedback for individuals with vision, hearing, or proprioceptive impairments [1] to providing an immersive experience by supporting Virtual Reality devices and elements of Virtual Environments [2]. Mechanoreceptors in our skin respond to various haptic stimuli such as vibration, pressure or skin stretch to provide information about the perceived sensation. Multi-haptic devices can stimulate different types of receptors, providing more information about the haptic sensations delivered and enabling better perception accuracy.

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Integrating more than one type of actuation system in a device, especially in soft wearable form factors, can bring in several complexities in terms of perceptual and mechanical interference, as well as the ability to discern the different types of haptic sensations provided. This research aims to understand the perceptual interference in a soft goods-based wearable device, focusing on the interaction between compression and vibration haptics.

Prior work in this area done by Zook [3] shows that perceptual interference is an effect that is perceived when two stimuli are presented simultaneously (stretch and squeeze). However, the setup used for this study was a rigid frame consisting of motors to deliver the haptic sensations; this makes the device bulky and limited in application. Among the different haptic actuation techniques, vibrotactile and compression/squeeze are two of the most commonly used types of haptic actuation. Given that vibrotactile has been used in so many contexts, the individual physical perception of vibrotactile stimulation across different parts of the body has also been studied extensively, and it found that the fingertips and the forearm are the most sensitive parts of the body to detect Vibration cues [4]. Compressive actuators, on the other hand, is relatively understudied as a haptic sensation. Compression is more natural form of delivering haptics and has many applications ranging from mimicking social touch [5] to communicating information. Although compressive actuators and vibrotactile actuators have been studied individually, the question of how the presence of one actuation interferes with the perception thresholds of another still needs to be answered. Especially when these are integrated with soft goods for designing wearable haptic systems where the interaction effects are much more prominent.

2 SYSTEM DESIGN

Given the requirement of accommodating and testing more than one type of actuation system, the proposed overall system will be designed, as shown in Figure 1(a). It will be a forearm device consisting of multiple repositionable actuation units, capable of delivering vibrotactile and compression haptic feedback. These repositionable elements allow testing of different system combinations, including placement/ physical distance between two actuation units of the same type and between two types. To design the compression actuation system, Shape Memory Alloys (SMAs) were used. SMAs are one of the most commonly used actuators for designing compression haptics due to their smaller form factor, which reduces the bulk of the system, making it easier to integrate into wearable devices.

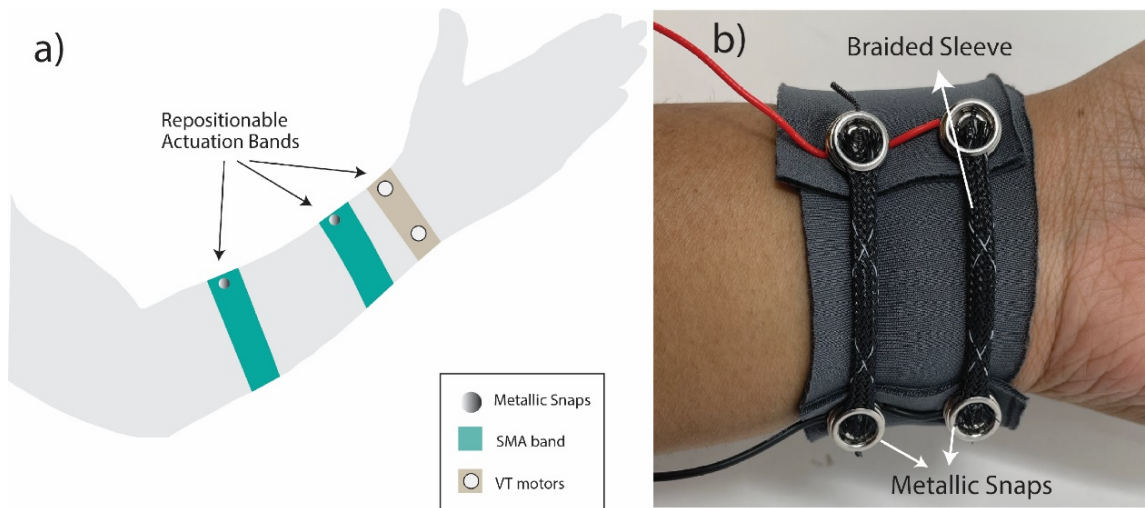


Figure 1: (a) System design consisting of repositionable SMA Compression and Vibrotactile actuation units for the forearm (b) Prototype of SMA actuator Unit

2.1 SMA Actuator Design

The design features of the SMA band focused on addressing three main areas of improvement: 1) Temperature insulation, 2) Incorporating wearability considerations and 3) Minimizing pressure points. Prior work has shown that heat generated by the SMA [6] and heat transfer are still major issues to address while designing SMA-based haptics.

The first version of the prototype, as seen in Figure 1 (b), uses a neoprene scuba knit (1.5mm thickness) as the base layer as well as the actuation layer due to the insulation and stretch properties of the material, with the SMA attached to the elastic using metallic snaps. In this design, Nitinol wires (Activation temperature of 70°C) made into springs of diameter (0.048in/0.121cm outer diameter), heat treated at 450°C, were used. The metallic snaps provide a means to replace the SMA in case of any damaged. To further insulate the SMA, 1/8th in diameter braided sleeves (Flexo flame retardant) were used. The base layer provided extra insulation and the change in skin temperature remained less than 7 °C when operated at 4.5W – 6W for a maximum of 10 seconds (measured using a thermal camera). This device was specifically designed to fit a single user based on the design guidelines by Granberry et al [7], although the stretch provided by the neoprene material would provide some allowances for users with dimensions closer to the test garment. Future iterations of this device will be designed to accommodate users within a specified size range which will help understand how variations in dimension affect perception. Compared with the prior work done in this space, this device is wearable, has increased insulation efforts with minimum layers, and consumes much less electrical power while still producing compression sensations.

3 FUTURE WORK

Future research work will involve prototype development of the vibrotactile unit design and validating the designs with user testing. To measure the effect of interference on perception, perceptual

quantities using the constant stimuli method will be used - the Just Noticeable Difference and the absolute threshold measurements. The experimental setup for this will focus only on administering compression and vibrotactile at a fixed distance. The baseline threshold values for vibrotactile only, Compression only, and the combinations of vibrotactile and compression and vice versa will be tested. This pilot study will be used to inform the protocol and the effect size for a larger study on understanding interference effects on perception with respect to other parameters such as spatial distance and visual cues.

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